
DIFFERENTIAL FORMS

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We express differential forms in terms of tensors and index notation. The closest treatment seems to be [Winitzki \(2009\)](#), but we improve even upon that in terms of clarity, exact short definitions and explicit examples.

It seems the usual old-fashioned index-based tensor formalism described in [Čertík, O. et al. \(2025\)](#) is all that is needed, and the differential forms provide an equivalent and optional treatment for antisymmetric tensors, which might be sometimes useful, but fundamentally does not bring any new feature or machinery, exactly the same computations can be done using index-based tensors also. We show how to equivalently rewrite any differential form formula or computation using tensors.

1 Vectors, Forms and Tensors

Any vector $\mathbf{u}$ can be written as:

$$\mathbf{u} = u^i \vec{e}_i. \quad (1)$$

Any form $\mathbf{p}$ can be written as:

$$\mathbf{p} = p_i \tilde{e}^i. \quad (2)$$

The basis vectors are $\vec{e}_i$ and the basis forms are $\tilde{e}^i$.

Any rank 2 tensor $\mathbf{A}$ can be written as:

$$\mathbf{A} = A^{ij} \vec{e}_i \otimes \vec{e}_j. \quad (3)$$

Any rank 3 tensor $\mathbf{A}$ can be written as:

$$\mathbf{A} = A^{ijk} \vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k. \quad (4)$$

The tensor basis is $\vec{e}_i \otimes \vec{e}_j$ and $\vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k$. If we mix lower and upper indices:

$$\mathbf{A} = A^i_j \vec{e}_i \otimes \tilde{e}^j. \quad (5)$$

The vector basis and the dual (form) basis is related by a dot product:

$$\tilde{e}^i \cdot \vec{e}_j = \delta^i_j. \quad (6)$$

If the vector basis is a coordinate basis, then it can be written as:

$$\vec{e}_i = \frac{\partial \mathbf{r}}{\partial x^i} \equiv \frac{\partial}{\partial x^i}, \quad (7)$$

where the position vector $\mathbf{r}$ can be omitted, and the basis vector is treated as an operator on any function on the manifold (we will not use this notation). The corresponding form basis can be written as

$$\tilde{e}^i \equiv dx^i. \quad (8)$$

The form can be said to act on a vector to produce a number (definition of a dual space) as follows using a dot product:

$$\mathbf{p}(\mathbf{u}) \equiv \mathbf{p} \cdot \mathbf{u} = p_i \tilde{e}^i \cdot u^j \vec{e}_j = p_i u^j \tilde{e}^i \cdot \vec{e}_j = p_i u^j \delta^i_j = p_i u^i. \quad (9)$$

We will not use the $\mathbf{p}(\mathbf{u})$ notation, we will use the dot product directly, or simply contracting indices as $p_i u^i$. All other forms acting on vectors is defined in a similar way using a dot product, and in index notation it just becomes a contraction.

2 Index Notation

We omit writing the basis vectors, so equations from the previous sections would be just u^i , p_i , A^{ij} , A^{ijk} , A^i_j , $p_i u^i$, etc. In most cases that is enough to fully represent everything. When we want to represent the tensor itself, we include the basis functions too.

The upper and lower indices are treated equally, they are just contravariant and covariant components of the same tensor.

3 Antisymmetric Tensors

An antisymmetric part of a tensor can be written as follows for rank 2:

$$A_{[ij]} = \frac{1}{2!} \varepsilon_{ij} \varepsilon^{kl} A_{kl} = \frac{1}{2} \det \begin{pmatrix} \delta_i^k & \delta_i^l \\ \delta_j^k & \delta_j^l \end{pmatrix} A_{kl} = \frac{1}{2} (\delta_i^k \delta_j^l - \delta_i^l \delta_j^k) A_{kl} = \quad (10)$$

$$= \frac{1}{2} (A_{ij} - A_{ji}). \quad (11)$$

Rank 3:

$$A_{[ijk]} = \frac{1}{3!} \varepsilon_{ijk} \varepsilon^{lmn} A_{lmn} = \frac{1}{3!} \det \begin{pmatrix} \delta_i^l & \delta_i^m & \delta_i^n \\ \delta_j^l & \delta_j^m & \delta_j^n \\ \delta_k^l & \delta_k^m & \delta_k^n \end{pmatrix} A_{lmn} = \quad (12)$$

$$= \frac{1}{3!} (\delta_i^l \delta_j^m \delta_k^n - \delta_i^m \delta_j^l \delta_k^n + \dots) A_{lmn} = \quad (13)$$

$$= \frac{1}{6} (A_{ijk} - A_{jik} + \dots - \dots + \dots). \quad (14)$$

Rank 4:

$$A_{[ijkl]} = \frac{1}{4!} \varepsilon_{ijkl} \varepsilon^{mnop} A_{mnop} = \frac{1}{4!} \det \begin{pmatrix} \delta_i^m & \delta_i^n & \delta_i^o & \delta_i^p \\ \delta_j^m & \delta_j^n & \delta_j^o & \delta_j^p \\ \delta_k^m & \delta_k^n & \delta_k^o & \delta_k^p \\ \delta_l^m & \delta_l^n & \delta_l^o & \delta_l^p \end{pmatrix} A_{mnop} = \quad (15)$$

$$= \frac{1}{4!} (\delta_i^m \delta_j^n \delta_k^o \delta_l^p - \delta_i^n \delta_j^m \delta_k^o \delta_l^p + \dots) A_{mnop} = \quad (16)$$

$$= \frac{1}{24} (A_{ijkl} - A_{jikl} + \dots). \quad (17)$$

The fraction at the beginning tells us how many terms there are in the final answer: 2, 6, 24 for ranks 2, 3, 4.

Note

The above expressions work for any dimension, except the Levi-Civita expression that only works for rank equal to dimension (e.g., rank 2, dimension 2). For dimension higher than the rank, one can skip the Levi-Civita expression and go straight to the determinant, which is correct in any dimension.

There is a dimension-dependent expression using Levi-Civita symbols that is equal to the determinant. For example for rank 3 and dimension 4, the expression is $\varepsilon_{ijks} \varepsilon^{slmn}$.

4 Exterior (Wedge) Product

The exterior (wedge) product of two forms $\mathbf{u} = u_i \tilde{e}^i$ and $\mathbf{v} = v_j \tilde{e}^j$ is defined as:

$$\mathbf{u} \wedge \mathbf{v} \equiv \mathbf{u} \otimes \mathbf{v} - \mathbf{v} \otimes \mathbf{u}. \quad (18)$$

Expressing this in components in terms of the basis:

$$\mathbf{u} \wedge \mathbf{v} = u_i \tilde{e}^i \wedge v_j \tilde{e}^j = \mathbf{u} \otimes \mathbf{v} - \mathbf{v} \otimes \mathbf{u} = u_i \tilde{e}^i \otimes v_j \tilde{e}^j - v_i \tilde{e}^i \otimes u_j \tilde{e}^j = \quad (19)$$

$$= u_i v_j \tilde{e}^i \otimes \tilde{e}^j - v_i u_j \tilde{e}^i \otimes \tilde{e}^j = \quad (20)$$

$$= (u_i v_j - u_j v_i) \tilde{e}^i \otimes \tilde{e}^j = \quad (21)$$

$$= \det \begin{pmatrix} u_i & u_j \\ v_i & v_j \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j = A_{ij} \tilde{e}^i \otimes \tilde{e}^j. \quad (22)$$

The tensor A_{ij} is antisymmetric. The last equation can be written using a wedge product by first rewriting the tensor basis into its symmetric and antisymmetric parts:

$$A_{ij} \tilde{e}^i \otimes \tilde{e}^j = A_{ij} \frac{1}{2} (\tilde{e}^i \otimes \tilde{e}^j + \tilde{e}^j \otimes \tilde{e}^i) + A_{ij} \frac{1}{2} (\tilde{e}^i \otimes \tilde{e}^j - \tilde{e}^j \otimes \tilde{e}^i) = \quad (23)$$

$$= A_{ij} \frac{1}{2} (\tilde{e}^i \otimes \tilde{e}^j + \tilde{e}^j \otimes \tilde{e}^i) + A_{ij} \frac{1}{2} \tilde{e}^i \wedge \tilde{e}^j = \frac{1}{2} A_{ij} \tilde{e}^i \wedge \tilde{e}^j. \quad (24)$$

The left term contains an antisymmetric tensor $A_{ij} = u_i v_j - u_j v_i$ contracted with a symmetric tensor, the result is zero. We end up with:

$$u_i \tilde{e}^i \wedge v_j \tilde{e}^j = (u_i v_j - u_j v_i) \tilde{e}^i \otimes \tilde{e}^j = \frac{1}{2} (u_i v_j - u_j v_i) \tilde{e}^i \wedge \tilde{e}^j. \quad (25)$$

Note

The equation

$$A_{ij} \tilde{e}^i \otimes \tilde{e}^j = \frac{1}{2} A_{ij} \tilde{e}^i \wedge \tilde{e}^j \quad (26)$$

shows that any rank 2 antisymmetric tensor can be written in terms of a wedge product as a differential form. And the definition (18) shows that any differential 2-form can be written as an antisymmetric tensor. The same will be true for higher ranks too.

Thus differential forms are simply antisymmetric tensors.

Note

The $\frac{1}{2}$ factor ends up there due to double counting: if we expand the original, in this note we'll assume dimension 2 for simplicity (i.e., $i = 1, 2$):

$$A_{ij} \tilde{e}^i \otimes \tilde{e}^j = A_{12} \tilde{e}^1 \otimes \tilde{e}^2 + A_{21} \tilde{e}^2 \otimes \tilde{e}^1 + A_{11} \tilde{e}^1 \otimes \tilde{e}^1 + A_{22} \tilde{e}^2 \otimes \tilde{e}^2 = \quad (27)$$

$$= A_{12} \tilde{e}^1 \otimes \tilde{e}^2 - A_{12} \tilde{e}^2 \otimes \tilde{e}^1 = A_{12} (\tilde{e}^1 \otimes \tilde{e}^2 - \tilde{e}^2 \otimes \tilde{e}^1) = A_{12} (\tilde{e}^1 \wedge \tilde{e}^2). \quad (28)$$

Above we used the fact that $A_{11} = A_{22} = 0$ and $A_{21} = -A_{12}$ due to antisymmetry. Now we expand the wedge formula in a similar manner:

$$\frac{1}{2} A_{ij} \tilde{e}^i \wedge \tilde{e}^j = \frac{1}{2} (A_{12} \tilde{e}^1 \wedge \tilde{e}^2 + A_{21} \tilde{e}^2 \wedge \tilde{e}^1 + A_{11} \tilde{e}^1 \wedge \tilde{e}^1 + A_{22} \tilde{e}^2 \wedge \tilde{e}^2) = \quad (29)$$

$$= \frac{1}{2} (A_{12} \tilde{e}^1 \wedge \tilde{e}^2 - A_{12} \tilde{e}^2 \wedge \tilde{e}^1) = \frac{1}{2} A_{12} (\tilde{e}^1 \wedge \tilde{e}^2 - \tilde{e}^2 \wedge \tilde{e}^1) = \quad (30)$$

$$= \frac{1}{2} A_{12} (\tilde{e}^1 \wedge \tilde{e}^2 + \tilde{e}^1 \wedge \tilde{e}^2) = A_{12} (\tilde{e}^1 \wedge \tilde{e}^2). \quad (31)$$

The final answer is exactly the same in both cases: $A_{12} (\tilde{e}^1 \wedge \tilde{e}^2)$. We can see that the terms $\tilde{e}^1 \wedge \tilde{e}^2$ and $-\tilde{e}^2 \wedge \tilde{e}^1$ combined due to antisymmetry, canceling the $\frac{1}{2}$. When we use the full tensor basis $\tilde{e}^i \otimes \tilde{e}^j$ then no

terms combine, we will always have 4 terms in 2D for a general asymmetric tensor. The wedge product basis combines the (12) and (21) terms, double counting the element values. Another way to put it is:

$$\tilde{e}^1 \wedge \tilde{e}^2 = \frac{1}{2}(\tilde{e}^1 \wedge \tilde{e}^2 - \tilde{e}^2 \wedge \tilde{e}^1) = (\tilde{e}^1 \otimes \tilde{e}^2 - \tilde{e}^2 \otimes \tilde{e}^1). \quad (32)$$

Each wedge product contains two tensor products, thus requiring the $\frac{1}{2}$. Alternatively, for both wedge and tensor products the $\frac{1}{2}$ comes from:

$$A_{ij} = \frac{1}{2}(A_{ij} - A_{ji}) \quad (33)$$

For an antisymmetric tensor A_{ij} .

Thus we get:

$$u_i \tilde{e}^i \wedge v_j \tilde{e}^j = \det \begin{pmatrix} u_i & u_j \\ v_i & v_j \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j = A_{ij} \tilde{e}^i \otimes \tilde{e}^j = \frac{1}{2} A_{ij} \tilde{e}^i \wedge \tilde{e}^j. \quad (34)$$

The wedge product of three 1-forms is defined using:

$$\mathbf{u} \wedge \mathbf{v} \wedge \mathbf{w} \equiv \mathbf{u} \otimes \mathbf{v} \otimes \mathbf{w} - \mathbf{v} \otimes \mathbf{u} \otimes \mathbf{w} + \mathbf{w} \otimes \mathbf{u} \otimes \mathbf{v} - \mathbf{w} \otimes \mathbf{v} \otimes \mathbf{u} + \quad (35)$$

$$+ \mathbf{v} \otimes \mathbf{w} \otimes \mathbf{u} - \mathbf{u} \otimes \mathbf{w} \otimes \mathbf{v}. \quad (36)$$

Expressing this in components we get a rank 3 antisymmetric tensor:

$$u_i \tilde{e}^i \wedge v_j \tilde{e}^j \wedge w_k \tilde{e}^k = \det \begin{pmatrix} u_i & u_j & u_k \\ v_i & v_j & v_k \\ w_i & w_j & w_k \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j \otimes \tilde{e}^k = A_{ijk} \tilde{e}^i \otimes \tilde{e}^j \otimes \tilde{e}^k. \quad (37)$$

TODO

Likely this is true:

$$A_{ijk} \tilde{e}^i \otimes \tilde{e}^j \otimes \tilde{e}^k = \frac{1}{3!} A_{ijk} \tilde{e}^i \wedge \tilde{e}^j \wedge \tilde{e}^k. \quad (38)$$

But we need to prove it.

Higher-order products are defined in analogous manner (n wedge products is a rank n fully antisymmetric tensor). The exterior product can be defined for vectors as well. The components of these tensors are equal to:

$$A_{ij} = \det \begin{pmatrix} u_i & u_j \\ v_i & v_j \end{pmatrix} = u_i v_j - u_j v_i \quad (39)$$

and

$$A_{ijk} = \det \begin{pmatrix} u_i & u_j & u_k \\ v_i & v_j & v_k \\ w_i & w_j & w_k \end{pmatrix}. \quad (40)$$

The wedge product is equal to the tensor written using the tensor basis in equations (34) and (37). The same tensor in index notation is given by equations (39) and (40).

Using properties of a determinant, we can see that the wedge product is just a fully antisymmetric tensor.

TODO

Derive all properties of the wedge product using existing tensor machinery: antisymmetry, all the common relations from literature, etc.

TODO

Show that in 2D and 3D the wedge product is the oriented area and volume. Ideally using tensors first, since we only want to use that, and then show that the wedge product when written using tensors gives the area and volume exactly.

TODO

Derive integration formulas by converting to tensor formulation and a metric. The determinant takes care of the volume and becomes the usual square root of a determinant of the metric I think.

5 Examples

5.1 Example: $dx \wedge dz$

We can now use the definitions to compute various special cases. For example if we have the exterior product of two forms dx and dz (which have components $dx_i = \delta_{1i}$ and $dz_i = \delta_{3i}$):

$$dx \wedge dz = \det \begin{pmatrix} \delta_{1i} & \delta_{1j} \\ \delta_{3i} & \delta_{3j} \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j = \tilde{e}^1 \otimes \tilde{e}^3 - \tilde{e}^3 \otimes \tilde{e}^1, \quad (41)$$

thus the components of this rank 2 tensor are:

$$A_{ij} = \det \begin{pmatrix} \delta_{1i} & \delta_{1j} \\ \delta_{3i} & \delta_{3j} \end{pmatrix} = \delta_{1i}\delta_{3j} - \delta_{1j}\delta_{3i} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}. \quad (42)$$

5.2 Example: $\tilde{e}^1 \wedge \tilde{e}^3$

The exterior product of two basis forms $\tilde{e}^1$ and $\tilde{e}^3$ is:

$$\tilde{e}^1 \wedge \tilde{e}^3 = \det \begin{pmatrix} \delta_{1i} & \delta_{1j} \\ \delta_{3i} & \delta_{3j} \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j = \tilde{e}^1 \otimes \tilde{e}^3 - \tilde{e}^3 \otimes \tilde{e}^1, \quad (43)$$

thus the components of this rank 2 tensor are:

$$A_{ij} = \det \begin{pmatrix} \delta_{1i} & \delta_{1j} \\ \delta_{3i} & \delta_{3j} \end{pmatrix} = \delta_{1i}\delta_{3j} - \delta_{1j}\delta_{3i} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}. \quad (44)$$

We can see that $\tilde{e}^1 \wedge \tilde{e}^3 = \tilde{e}^1 \otimes \tilde{e}^3 - \tilde{e}^3 \otimes \tilde{e}^1$, so the wedge product is simply an antisymmetric tensor, written with the form basis functions included.

5.3 Example: $dx^1 \wedge dx^2 \wedge dx^4$ evaluated on vectors

When evaluating the form (composed of basis forms) on vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$, we are computing the following dot product of the two rank 3 tensors (full contraction):

$$(dx^1 \wedge dx^2 \wedge dx^4) \cdot (\mathbf{u} \otimes \mathbf{v} \otimes \mathbf{w}) = \quad (45)$$

$$= \det \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{4i} & \delta_{4j} & \delta_{4k} \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j \otimes \tilde{e}^k \cdot (u^l v^m w^n \tilde{e}_l \otimes \tilde{e}_m \otimes \tilde{e}_n) = \quad (46)$$

$$= \det \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{4i} & \delta_{4j} & \delta_{4k} \end{pmatrix} u^i v^j w^k = \det \begin{pmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_4 & v_4 & w_4 \end{pmatrix}. \quad (47)$$

The form has thus selected the first, second and fourth component row of the vectors. The result is just a scalar.

6 Exterior Derivative

The exterior derivative d is defined using the following rules / definitions for differential forms:

- 0-form f : $df \equiv \frac{\partial f}{\partial x^i} dx^i = \partial_i f dx^i$
- k-forms α and β , numbers a, b : $d(a\alpha + b\beta) = ad\alpha + bd\beta$
- k-form α , l-form β : $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$
- k-form α : $d(d\alpha) = 0$

It turns out in terms of tensors, the exterior derivative is simply a regular derivative that is made antisymmetric:

- 0-form: $\partial_i f$
- 1-form: $\partial_{[i} f_{j]}$
- 2-form: $\partial_{[i} A_{jk]}$
- 3-form: $\partial_{[i} A_{jkl]} = 0$ (in 3 dimensions, since rank 4 fully antisymmetric tensor in 3D is identically 0)

One can use the formulas from the section “Antisymmetric Tensors” above to evaluate these antisymmetric derivatives. We show examples of this below. We first compute the exterior derivative using differential forms and the above rules, and then we do exactly the same operations using tensors.

6.1 Differentiating 0-forms

Input:

$$f(x, y). \quad (48)$$

Derivative:

$$df = \partial_x f dx + \partial_y f dy. \quad (49)$$

In general, we get:

$$df = \partial_i f dx^i. \quad (50)$$

6.2 Differentiating 1-forms

Input:

$$\omega = f_1(x^1, x^2) dx^1 + f_2(x^1, x^2) dx^2. \quad (51)$$

Derivative:

$$d\omega = df_1 \wedge dx^1 + df_2 \wedge dx^2 = \partial_i f_1 dx^i \wedge dx^1 + \partial_i f_2 dx^i \wedge dx^2 = \quad (52)$$

$$= \partial_2 f_1 dx^2 \wedge dx^1 + \partial_1 f_2 dx^1 \wedge dx^2 = (\partial_1 f_2 - \partial_2 f_1) dx^1 \wedge dx^2. \quad (53)$$

In general, the derivative of a 1-form $\tilde{f} = f_i dx^i$ is:

$$d\tilde{f} = d(f_i dx^i) = df_i \wedge dx^i = (\partial_j f_i dx^j) \wedge dx^i = \frac{1}{2} (\partial_j f_i - \partial_i f_j) dx^j \wedge dx^i = \quad (54)$$

$$= (\partial_j f_i - \partial_i f_j) dx^j \otimes dx^i. \quad (55)$$

So we get:

$$d\tilde{f} = A_{ij} dx^i \otimes dx^j, \quad (56)$$

where $A_{ij} = \partial_i f_j - \partial_j f_i$.

We take a 1-form f_j then we differentiate $\partial_i f_j$, this is a rank 2 tensor, and we project it into the antisymmetric subspace $dx^i \otimes dx^j - dx^j \otimes dx^i = \frac{1}{2} dx^i \wedge dx^j$, which cancels the symmetric part of the derivative and only the antisymmetric part survives: $\partial_i f_j - \partial_j f_i$. So we define the derivative to be $A_{ij} = \partial_i f_j - \partial_j f_i$. We can define the derivative on antisymmetric tensors (exterior derivative) to be a regular derivative projected into the antisymmetric subset.

Let's now do the same computation using regular tensors:

$$A_{ij} = \partial_{[i} f_{j]} = \frac{1}{2} \varepsilon_{ij} \varepsilon^{kl} \partial_k f_l = \frac{1}{2} (\delta_i^k \delta_j^l - \delta_i^l \delta_j^k) \partial_k f_l = \frac{1}{2} (\partial_i f_j - \partial_j f_i). \quad (57)$$

The exterior derivative is simply a regular (not antisymmetric) derivative $\partial_i f_j$ that is made antisymmetric: $\partial_{[i} f_{j]}$. The same applies to higher ranks.

6.3 Differentiating 2-forms in 3D

Input:

$$\omega = F^1 dx^2 \wedge dx^3 + F^2 dx^3 \wedge dx^1 + F^3 dx^1 \wedge dx^2. \quad (58)$$

Output:

$$d\omega = dF^1 \wedge dx^2 \wedge dx^3 + dF^2 \wedge dx^3 \wedge dx^1 + dF^3 \wedge dx^1 \wedge dx^2 = \quad (59)$$

$$= \partial_1 F^1 dx^1 \wedge dx^2 \wedge dx^3 + \partial_2 F^2 dx^2 \wedge dx^3 \wedge dx^1 + \partial_3 F^3 dx^3 \wedge dx^1 \wedge dx^2 = \quad (60)$$

$$= (\partial_1 F^1 + \partial_2 F^2 + \partial_3 F^3) dx^1 \wedge dx^2 \wedge dx^3. \quad (61)$$

In general for antisymmetric A_{ij} :

$$\omega = A_{ij} dx^i \otimes dx^j = \frac{1}{2} A_{ij} dx^i \wedge dx^j. \quad (62)$$

We differentiate the tensor:

$$d\omega = \frac{1}{2} dA_{ij} dx^i \wedge dx^j = \frac{1}{2} \partial_k A_{ij} dx^k \wedge dx^i \wedge dx^j = \quad (63)$$

$$= \frac{1}{2} \partial_{[k} A_{ij]} dx^i \wedge dx^j \wedge dx^k = \quad (64)$$

$$= \frac{1}{2} \frac{1}{6} \varepsilon_{kij} \varepsilon^{lmn} \partial_l A_{mn} dx^i \wedge dx^j \wedge dx^k = \quad (65)$$

$$= \frac{1}{2} \frac{1}{6} \varepsilon_{kij} \varepsilon^{lmn} \partial_l \varepsilon_{mnq} F^q dx^i \wedge dx^j \wedge dx^k = \quad (66)$$

$$= \frac{1}{2} \frac{1}{6} \varepsilon_{kij} 2\delta^l_q \partial_l F^q dx^i \wedge dx^j \wedge dx^k = \quad (67)$$

$$= \frac{1}{6} \varepsilon_{ijk} \partial_q F^q dx^i \wedge dx^j \wedge dx^k = \quad (68)$$

$$= \partial_q F^q dx^1 \wedge dx^2 \wedge dx^3. \quad (69)$$

We used the fact that the basis is antisymmetric, so only the antisymmetric part of the derivative survives. The basis becomes antisymmetric due to the incorporation of the dx^k base via the wedge product, otherwise the derivative is not fully antisymmetric.

The above is the correct result for differentiating any k -form in n -D space where $k = n - 1$.

Let's now do the same computation using regular tensors:

$$\partial_{[k} A_{ij]} = \frac{1}{6} \varepsilon_{kij} \varepsilon^{lmn} \partial_l A_{mn} = \frac{1}{6} \varepsilon_{kij} \varepsilon^{lmn} \partial_l \varepsilon_{mnq} F^q = \quad (70)$$

$$= \frac{1}{6} \varepsilon_{kij} 2\delta^l_q \partial_l F^q = \frac{1}{6} \varepsilon_{ijk} \partial_q F^q. \quad (71)$$

The exterior derivative is simply a regular (not antisymmetric) derivative $\partial_k A_{ij}$ that is made antisymmetric: $\partial_{[k} A_{ij]}$.

7 TODO

TODO

Add examples of a 3D space, rank 0-, 1-, 2-, 3-forms that correspond to scalars, vectors/forms, rank 2 antisymmetric tensor (which internally contains a pseudo-vector I think, using $A_{ij} = \varepsilon_{ijk} B_k$; here ε_{ijk} is a tensor density, not a tensor, so also B_k is not a vector, but a pseudovector), and rank 3 antisymmetric tensor, which I think must contain only one independent component, and it is related to the oriented volume, the component might be a pseudoscalar, since it changes sign when you flip axes, or something like that.

Riemannian volume form ($\sqrt{|g|} dx^1 \wedge \dots \wedge dx^n$) is related to this, add examples.

$\sqrt{-g}$ is a scalar density, add examples.

Show this:

$$\mathbf{a} \wedge \mathbf{b} = i \mathbf{a} \times \mathbf{b}, \quad (72)$$

where $i = e_1 \wedge e_2 \wedge e_3$ is a unit pseudoscalar, $i^2 = -1$.

This is also related to the Hodge star operator that provides isomorphism between axial vectors (pseudovectors) and bivectors (2-forms).

Give examples from physics for each. It should be possible to build this structure fully using antisymmetric tensors.

Also add examples of differentiation, where derivative of a vector creates a tensor, this must be related to $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ which is antisymmetric, so this operation can be expressed as differentiating differential forms, since both LHS and RHS are antisymmetric tensors. Differentiating one more time in the antisymmetric manner then creates the rank 3 tensor. No more antisymmetric derivatives are then possible.

Knowing exactly where each physical entity belongs is crucial (give examples), but that is possible already using tensors, we do not need differential forms for that. Rewrite various equations that use differential forms to use tensors.

References

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